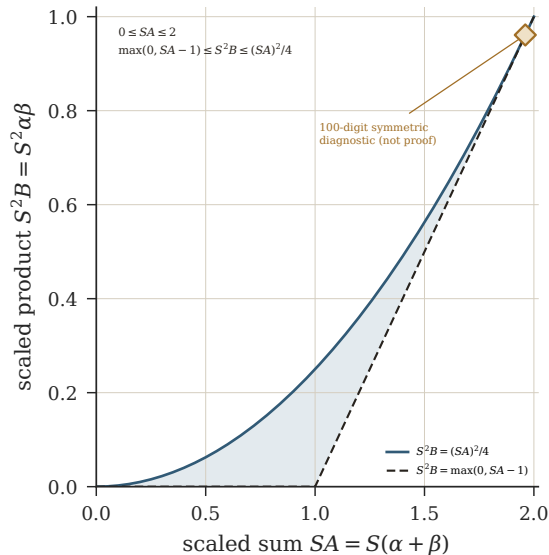


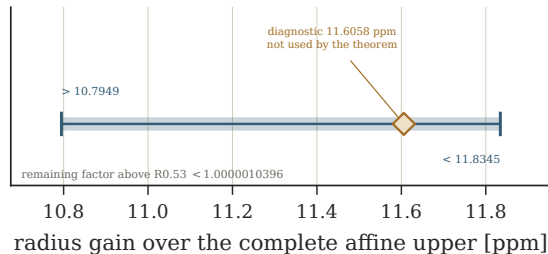


A global exact enclosure for the complete product-affine family

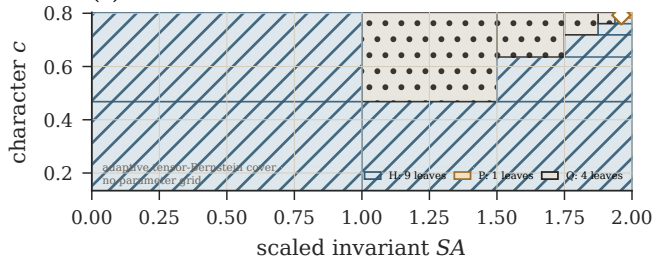
(a) Exact compactified invariant domain



(b) Certified gain interval for the whole family



(c) Exact 14-leaf cover at $r = 0.382629$



Theorem: $0.382628602237879637 < r_{\text{prod}}^{\text{opt}} < 0.382629$ in the complete compactified product-affine family of the degree-80 reduced edge model.

Proof: exact invariant reduction, character-tail exclusion, and a complete dyadic tensor-Bernstein cover. The plotted boundary samples and optimizer point are not proof data.

Scope: reduced algebraic model only · 16/16 exact checks · 14 leaves · GMP rational signs · no three-dimensional Navier-Stokes regularity claim